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Goal

- Enable battery-focused material exploration and candidate analysis for EV and energy storage applications.

Key Business Questions

- Which materials are lightweight and conductive?
- Which materials show promising battery characteristics?
- How do experimental and calculated band gaps compare?
- Which materials provide optimal performance/stability balance?

KEY INSIGHTS

Top Candidates Material Ranking

For battery cathode materials we actually want:

- Low density (lightweight)
- Moderate, non-zero band gap (semiconductors, not metals — metals are short-circuit risks)
- Low energy_above_hull (thermodynamically stable)

Conductivity vs. Density

Bottom-Left corner: These materials are the perfect candidates for high-performance batteries because they don't weigh much but allow electricity to move through them easily.

- X-Axis (Density): Further to the Left is better (lighter batteries).
- Y-Axis (Band Gap): Further to the Bottom is more conductive.

Experimental vs. Calculated Band Gaps

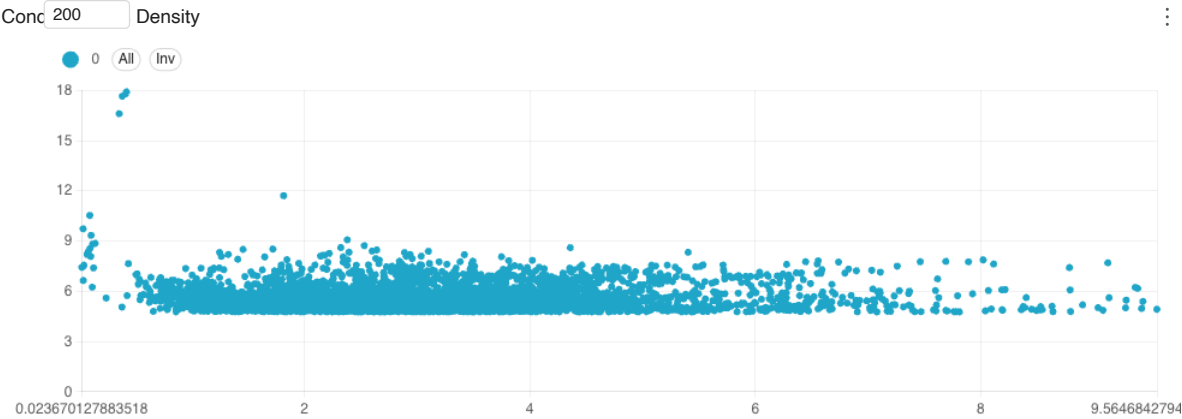
The Goal: Validating our "Digital Twin" models. Before we build a factory, we calculate the properties with AI/Physics.

- The Diagonal Line (\$y=x\$): If a dot sits on the diagonal, our calculations were perfect.

Material Similarity Explorer (LFP Comparison)

LFP (Lithium Iron Phosphate) is one of the most successful battery cathodes in the world (used by Tesla and others) because it is safe and cheap What to look for: Look for a material that has a Delta Stability of ~0 (just as safe as LFP) but a Delta Density that is negative (lighter than LFP).

Top Candidate Materials Ranking



Material Similarity Explorer - most similar material to LFP (Lithium Iron Phosphate)

material_id	density	band_gap	energy_above_hull	crystal_system	is_metal	delta_density	delta_band_gap	delta_stability
mp-19017			0	Orthorhombic	0	0	0	0
mp-29703			0	Monoclinic	0	0.0018	0.01	0
mp-1190479			0.0074	Monoclinic	0	0.0012	0.0088	0.0074
mp-1046445			0.0118	Triclinic	0	0	0.0045	0.0118
mp-1195392			0.0112	Monoclinic	0	0.0009	0.0047	0.0112
mp-626726			0.0034	Monoclinic	0	0.0154	0.0063	0.0034
mp-636598			0.002	Monoclinic	0	0.0264	0.0031	0.002
mp-570168			0.0054	Orthorhombic	0	0.0153	0.0099	0.0054
mp-1222575			0.0109	Monoclinic	0	0.0252	0.0003	0.0109
mp-1194662			0	Orthorhombic	0	0.0063	0.0391	0



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Battery Material Explorer

Experimental vs Calculated Band Gaps

